Why accurate neural predictions do not guarantee effective neural control

Summary. Deep encoding models of visual cortex can effectively predict natural-image responses while failing to generate synthetic images that control neural firing (Prince & Wang et al., 2026). Here we develop a regression theory to explain this dissociation. We consider linear regression from linear image features to noisy neural responses. We show that weight errors along low-variance directions of data matter little for accurate natural-image prediction but can severely distort predictions for synthetic images optimized through backpropagation to pixel space. Using random matrix theory, we then show how limited data and noise can cause models to overestimate how strongly these images will drive neurons. Regularization tuned for natural-image prediction need not prevent this overprediction. Control also depends strongly on the model's feature representation: we illustrate how different feature spaces can support equally accurate predictions for natural images while guiding synthesis toward images with very different control efficacy. The theory yields a gradient-geometry measure that predicts empirical control outcomes better than natural-image prediction accuracy. Overall, these results explain why natural-image prediction need not translate into effective control and suggest concrete routes to improvement through more data, reduced noise, and model selection guided by feature-gradient geometry.

The experimental puzzle. Prince, Wang, et al. [1] recorded neuronal responses to naturalistic images at 25 sites in five macaques, spanning V3/V4, inferotemporal cortex (IT), and superior temporal sulcus (STS). For each site, ridge regression fitted linear readouts from hidden-layer features of ten fixed deep networks, with the feature layer selected by cross-validation. This yielded ten encoding models per site, $f_m(x) = \hat{w}_m^\top \phi_m(x)$. Feature accentuation [2] then generated images that each encoding model predicted would evoke a target response r^*. Starting from a seed image, regularized gradient ascent or descent on f_m modified the image until its predicted response reached the target (schematically, $x \leftarrow x \pm \eta \nabla_x f_m(x)$). Repeating this procedure for targets spanning the neuron's natural-image response range produced an accentuation sweep (Fig. 1A,B). The generated images were presented to the same sites in a second recording session, testing whether measured responses matched the model's targets and thus assessing model-based control. Although trained models achieved similar natural-image predictivity, their control performance differed sharply (Fig. 1E,F): an accurate predictor could barely modulate its target site (Fig. 1B–D). Some unsuccessful controllers could nevertheless predict responses to effective stimuli generated by another model.

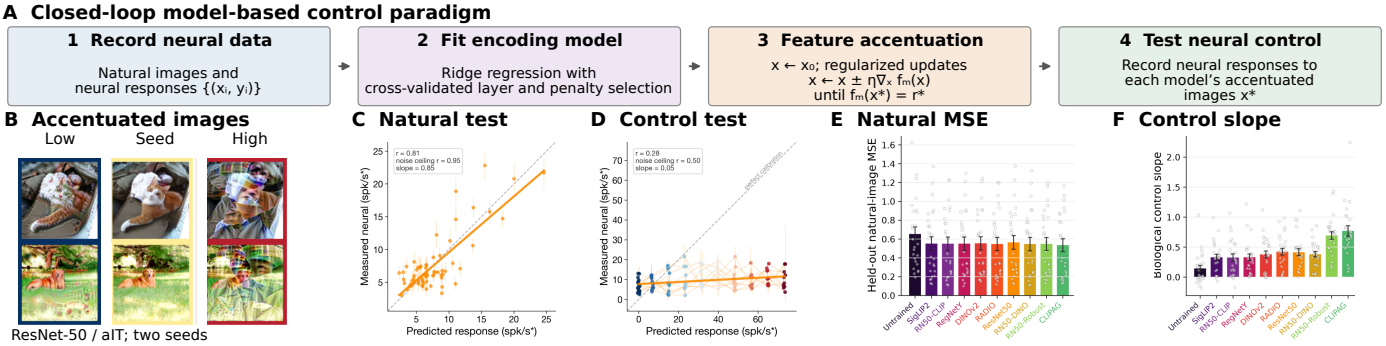


Figure 1. From encoding to control. A Experimental paradigm of closed-loop model-based control comparison. **B–D** One ResNet-50/aIT site from [1]: two accentuation sweeps; natural test responses ($r = 0.81$, slope 0.85); control responses ($r = 0.28$, slope 0.05). **E,F** Natural-image MSE and control slope across ten models: sites (dots), mean and ± 1 SE (bars); 25 sites.

From experiment to a solvable model. We preserve the backbone–readout structure but replace $\phi(x)$ by fixed linear features Fx. Natural inputs $x \sim \mathcal{N}(0, \Sigma)$ drive a noisy linear neuron, $y = x^\top \beta^\star + \sigma\epsilon$, with independent unit-variance noise ϵ and signal variance $S = \beta^{\star\top}\Sigma\beta^\star$. Ridge regression on n examples fits $f(x) = \hat{w}^\top Fx = x^\top \hat{\beta}$, with pixel gradient $\hat{\beta} = F^\top \hat{w}$ (Fig. 2A). We retain ridge fitting but idealize regularized synthesis as the straight path $x = x_0 + \alpha\hat{\beta}$. With $x_0 = 0$ and a centered sweep $\mathbb{E}[\alpha] = 0$, mean neural and predicted response changes are αN and αD:

$$\text{slope}_{\text{control}} = \frac{N}{D} = \frac{\hat{\beta}^\top \beta^\star}{\|\hat{\beta}\|^2} \begin{matrix} \text{actual neural gain} \\ \text{model-predicted gain} \end{matrix}, \qquad R^2_{\text{control}} = 1 - \left(\frac{D}{N} - 1\right)^2. \tag{1}$$

Alignment N is the actual neural gain; squared norm D is the model's claimed gain. For $N > 0$, $D < N$ means underprediction (slope > 1), $D = N$ means perfect calibration, and $D > N$ means overprediction (slope < 1), which could lead to negative R^2 (when $D > 2N$). Natural-image prediction instead measures $E_{\text{gen}} = (\hat{\beta} - \beta^\star)^\top \Sigma (\hat{\beta} - \beta^\star)$, downweighting low-variance directions. Errors along these directions can therefore inflate the model's claimed gain D with little predictive cost or corresponding neural gain N.

Why cross-validation does not remove the problem. The failure already occurs for $F = I$, with no backbone geometry. Random matrix theory [3] gives leading approximations as dimension and sample size grow

proportionally. The mean weight is the teacher shrunk along each covariance eigenvector:

$$\bar{\beta} := \Sigma(\Sigma + \kappa I)^{-1}\beta^\star = \sum_k \frac{s_k}{s_k + \kappa} b_k u_k, \qquad \Sigma u_k = s_k u_k, \quad b_k = u_k^\top \beta^\star. \tag{2}$$

Here κ is the effective ridge penalty accounting for finite sampling; each shrinkage factor $s_k/(s_k+\kappa) \in [0,1]$. Finite-data fluctuations inflate squared norm D by V, without changing leading alignment N:

$$\text{slope}_{\text{control}} \simeq \frac{\bar{\beta}^\top \beta^\star}{\|\bar{\beta}\|^2 + V}, \qquad V = \frac{E_{\text{gen}} + \sigma^2}{n} \sum_k \frac{s_k}{(s_k+\kappa)^2}, \tag{3}$$

For the mean alone, $\|\bar{\beta}\|^2 \le \bar{\beta}^\top \beta^\star$: the student underpredicts. Inflation increases the model-predicted gain D without increasing the neural gain N. It first compensates for shrinkage, giving $D = N$, then causes overprediction once $D > N$. For independent ridge fits, V drops out of the peer weight cross-product, yielding shrinkage-driven underprediction, consistent with neural data (Fig. 2E).

Prediction-optimal and control-calibrated penalties satisfy different spectral conditions for anisotropic inputs and a structured teacher. Natural-image simulations exhibit distinct optima (Fig. 2B): selecting a penalty for prediction need not calibrate control.

Why equally predictive feature maps control differently. For general F, prediction depends on $F\Sigma F^\top$ and $h = F\Sigma\beta^\star$, with signal variance, noise and sample size fixed; control also depends on $FF^\top$ and $F\beta^\star$. We construct transformations preserving the joint feature–response distribution, hence the predicted generalization curve and its optimal penalty stay invariant, while control changes (Fig. 2C). Increasing input anisotropy can make this dissociation arbitrarily severe.

The feature-gradient geometry enters the norm inflation,

$$V_F = \frac{E_{\text{gen}} + \sigma^2}{n} T_F(\kappa), \qquad T_F(\kappa) = \sum_{k:s_k>0} \left(\frac{s_k}{s_k+\kappa}\right)^2 \frac{\|g_k\|^2}{g_k^\top \Sigma g_k}, \tag{4}$$

where (s_k, v_k) now denote feature-covariance eigenpairs and $g_k = F^\top v_k$ is a feature direction's pixel gradient. The sum weights features that survive regularization and compares their gradient energy with natural-image variation along that gradient. Thus, control is fragile when well-estimated features send their gradients into weakly constrained pixel directions, making T_F large without changing prediction error.

From linear theory to biological control. For nonlinear features ϕ, the feature Jacobian J_ϕ at seed images gives $T_\phi = \sum_k s_k/(s_k+\kappa)^2 \|J_\phi(x)^\top v_k\|^2$. Averaging squared gradients over a Gaussian input neighborhood (standard deviation $16/255$) and scaling by held-out mean squared error (MSE) divided by n gives a neighborhood inflation proxy V_ϕ. Across nine trained models and 25 sites (225 pairs), raw $\log_{10}$ predictors correlate with site-centered control slope at $r = -0.61$ for T_ϕ, -0.54 for neighborhood V_ϕ, and 0.02 for held-out MSE (Fig. 2D). Site means are computed over all ten models before excluding the untrained baseline. These descriptive associations support assessing gradient geometry alongside prediction accuracy.

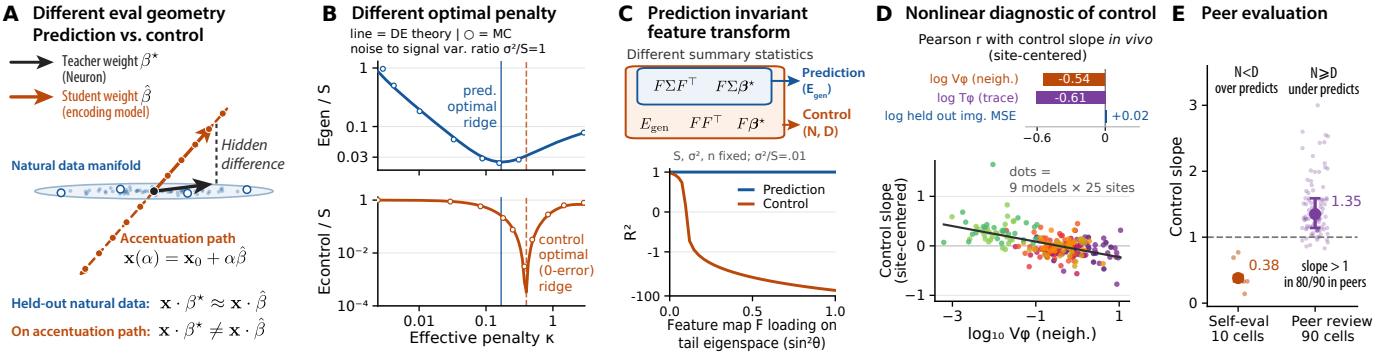


Figure 2. Prediction–control dissociation. **A** Weight errors along low-variance data modes exposed by accentuation. **B** Distinct optimal ridge penalties. **C** Prediction and control have different summary statistics; Feature transform can change control without touching prediction. **D** Log-predictor correlations with site-centered slope (nine models, 25 sites). **E** Site-averaged self/peer slopes; bars: median and interquartile range.

Toward more reliable neural control. The factorization $V_F = (E_{\text{gen}} + \sigma^2)T_F/n$ turns the failure mechanism into a design guide: improve predictive fits, collect more training images, and average trials or denoise responses. At fixed effective penalty, T_F adds a distinct criterion: among comparably predictive backbones, favor or regularize gradients that place less energy in low-variance image directions. The nonlinear diagnostic makes this criterion testable before synthesis. Reducing inflation addresses overprediction; calibration still requires balancing it against shrinkage. The design target is calibrated gain: optimized images should change neural responses by as much as the model predicts.

References. [1] Prince, Wang, et al. (2026). Parametric neural control differentiates top neural network models of primate visual cortex. *bioRxiv*, doi:10.64898/2026.08.16.745063. [2] Hamblin et al. (2024). Feature accentuation: Revealing 'what' features respond to in natural images. arXiv:2402.10039. [3] Dobriban & Wager (2018). High-dimensional asymptotics of prediction: Ridge regression and classification. *Annals of Statistics* 46:247–279.